LITERATURE REVIEW

Global Factors in Non-core Bank Funding and Exchange Rate Flexibility

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ANALYTICAL ABSTRACT

The paper asks whether the share of bank funding that is not deposit based, measured as foreign liabilities over deposits (the non-core ratio), is mostly driven by global rather than national forces, and whether a flexible exchange rate shields a banking system from those forces. Using a quarterly panel of 30 advanced economies from 2004 to 2022, the authors estimate a dynamic panel with unobserved common factors in three steps: a common correlated effects mean group regression, principal components of the resulting composite error, and a final regression in which the components enter with loadings that differ between pegged and floating economies. Three components account for about 80 percent of the explained common variation, are tied to world macroprudential tightness, the US shadow rate and crisis episodes, and load more heavily on fixed exchange rate countries, mostly through interbank liabilities. Tighter macroprudential policy appears to help fixed regimes. The central limitation is that the fixed regime group is almost entirely euro area members, so the exchange rate regime is confounded with monetary union and with cross-border interbank integration, and several headline differences are not individually significant.

1 Introduction

1.1 Paper identification and classification

The paper is an empirical and econometric study in international finance, at the intersection of the global financial cycle literature, the trilemma literature and the literature on bank funding structure. The subfield is international banking: the transmission of global financial conditions to national banking systems and the role of the exchange rate regime in that transmission. The version analysed is the arXiv HTML version v5, read from the abstract to the data appendix and the list of figures. Text, equations, all numbered tables and the appendix tables were available. The six figures were present only as captions, so the plotted series could not be inspected and the analysis of figures relies on the captions and on the underlying tables. The date printed on the HTML version is a compilation date and is not used here. The arXiv record lists a later version v6 (April 2025) that was not read, so any change between v5 and v6 is outside this review.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

PAPER The paper tests two linked propositions. The first is that fluctuations in the ratio of non-core to core bank funding in advanced economies are explained mainly by a small number of global factors, with country specific factors playing a minor role. The second is that exchange rate flexibility reduces the sensitivity of that ratio to those global factors, which is the banking balance sheet version of the Mundell trilemma.

WHY DOES THE QUESTION MATTER?

INFERENCE Non-core funding is more elastic and more run prone than deposits. When it is foreign and short term it ties domestic credit supply to the lending capacity of global banks and thus to conditions abroad. If a floating currency lets the national banking system decouple from foreign funding conditions, then the case for monetary policy independence under free capital mobility survives. If it does not, as Rey (2015) argued, the relevant policy menu shifts towards capital flow management and macroprudential tools. The balance sheet ratio studied here is also a crisis predictor in the long historical panels of Jordà, Schularick and Taylor (2017), so the question has a stability dimension as well as a transmission dimension.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

PAPER The authors cite one strand that finds a dominant global factor in capital flows and credit (Rey 2015; Miranda-Agrippino and Rey 2020; Passari and Rey 2015; Amiti, McGuire and Weinstein 2017) and another that finds partial insulation from exchange rate flexibility and from prudential tools (Shambaugh 2004; Obstfeld, Shambaugh and Taylor 2005; Obstfeld, Ostry and Qureshi 2019; Aizenman, Chinn and Ito 2016). Cerutti, Claessens and Rose (2019) find that the global component of

capital flows explains less than a quarter of their variance. CRITIQUE What is missing, in the authors' framing, is a treatment of cross-country dependence that is econometrically adequate: ordinary two-way fixed effects do not remove strong cross-sectional dependence when loadings on global factors differ across countries, which is exactly the heterogeneity the trilemma is about.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

PAPER The paper proposes a three-step estimator: common correlated effects mean group (CCE-MG) estimation, extraction of principal components from the composite residual, and a final mean group regression with component loadings split by exchange rate regime and by macroprudential stance. It finds three components, which together carry about 80 percent of the explained common variation. The first correlates with world macroprudential tightness, the second with the US shadow rate and the third with crisis dummies. Loadings are larger for fixed regimes, the difference is stated to be stronger outside crises, and stricter macroprudential policy reduces loadings on the second and third components, mainly for fixers. The mechanism is traced to interbank liabilities.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether, in a balanced quarterly panel of advanced economies, the common (global) component of the ratio of foreign liabilities to deposits is larger for banking systems under strictly fixed exchange rate regimes than for those under floating regimes, after removing cross-sectional dependence through common correlated effects and principal components. A secondary question is whether the same split by tight versus loose cumulative macroprudential policy produces a similar contrast.

2.2 Hypotheses

- H1, global dominance: most variation in the non-core ratio is common across countries and is captured by about three factors, with domestic variables playing a small role.
- H2, trilemma insulation: loadings on the global components are smaller for floating than for fixed regimes.
- H3, crisis dependence: the insulation of H2 is larger in non-crisis periods, because the third component, associated with crises, is not insulated.
- H4, prudential complement: tighter macroprudential policy reduces loadings, especially for fixed regimes.
- H5, channel: the effect operates through interbank cross-border liabilities, independent of the currency denomination.

2.3 Existing literature and research gap

PAPER The authors position the work against the global financial cycle (Rey 2015), the bank leverage and global liquidity literature (Shin and Shin 2011; Bruno and Shin 2015; Hahm, Shin and Shin 2013) and the panel econometrics of common factors (Pesaran 2006; Bai 2009; Chudik and Pesaran 2015). CRITIQUE The gap is real in method, since earlier work on the trilemma often used two-way fixed effects or single global proxies such as the VIX, but the paper does not show that earlier conclusions would have been reversed by correct treatment of cross-sectional dependence. Table 2 in fact shows the two-way fixed effects columns agreeing in sign with the final estimates for the real exchange rate and the money to GDP ratio, so the methodological correction matters mostly for the global factor structure, not for the domestic controls.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: identification of the global factor structure of non-core funding, evidence that exchange rate flexibility insulates, and evidence that macroprudential policy complements it. The first is demonstrated at the level of statistical decomposition, but the economic identity of the factors rests on correlations in Table 6, not on structural identification. The second is supported in sign but not in sharp statistical terms (Section 5 below), and its interpretation is confounded by the composition of the fixed group. The third rests on country groups of six and ten economies. The contribution is therefore methodological and descriptive rather than causal.

3 Theoretical Framework and Mechanism

3.1 Economic mechanism

PAPER The paper has no formal economic model. The mechanism is imported from the global banking literature: bank equity is sticky and deposits are inelastic over the cycle, so credit expansion has to be funded with non-core liabilities, of which foreign borrowing is a large component (Shin and Shin 2011; Hahm, Shin and Shin 2013; Bruno and Shin 2015). The ratio of foreign liabilities to deposits therefore rises when global banks have lending capacity and when domestic credit demand is strong, and falls in global deleveraging. The exchange rate regime enters as follows. INFERENCE Under a peg the central bank imports foreign monetary conditions, so a looser global stance transmits directly to domestic funding costs and to the incentive to borrow abroad. Under a float the domestic rate can move away from the foreign one and the exchange rate absorbs part of the shock, which should reduce the arbitrage incentive to borrow abroad and the balance sheet effects of currency valuation on bank leverage.

3.2 The econometric model

PAPER The analysis starts from an interactive fixed effects model with unit specific slopes. For country i and quarter t, with N countries and T quarters,

$$y_{i,t} = \rho_i y_{i,t-1} + x_{i,t}\beta_i + g_t\delta_i + u_{i,t}, \qquad u_{i,t} = f_t\gamma_i + \varepsilon_{i,t}. \tag{1}$$

- $y_{i,t}$ non-core ratio, foreign liabilities over transferable and other deposits, in percent.
- ρ_i country specific persistence, so that heterogeneous dynamics are allowed.
- $x_{i,t}$ $1 \times K$ vector of strictly exogenous country variables: GDP growth, real interest rate, real effective exchange rate, capital to asset ratio, capital and liquidity regulation, money to GDP.
- g_t M_{ocf} observed common factors: VIX, oil price, US shadow rate, world non-bank share, world GDP growth, world macroprudential tightness, world inflation and the US real effective exchange rate.
- f_t M_{ucf} unobserved common factors, with country loadings γ_i.
- $\varepsilon_{i,t}$ idiosyncratic noise, assumed free of cross-sectional dependence.

Equation (1) says that every country is hit by the same $M_{ocf} + M_{ucf}$ time series, but each country responds with its own loading. A two-way fixed effects estimator removes only a common time mean, which equals $\bar{\gamma} f_t$, and leaves $(\gamma_i - \bar{\gamma}) f_t$ in the error. This remainder is correlated with $x_{i,t}$ whenever the regressors themselves load on the same factors:

$$x_{i,t} = g_t\delta_i^x + f_t\gamma_i^x + \xi_{i,t}. \tag{2}$$

The consequence is omitted variable bias and, because the omitted term has a non-vanishing cross-sectional covariance, inconsistent estimates of β_i. Strong cross-sectional dependence is tested with the Pesaran (2015) statistic

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}, \tag{3}$$

where $\hat{\rho}_{ij}$ is the pairwise correlation of residuals. Under weak dependence CD is asymptotically standard normal; a large absolute value signals leftover common factors. **INFERENCE** The statistic is the average pairwise correlation scaled by $\sqrt{TN(N-1)/2}$, so even a small average correlation produces a large CD in a panel of this size, which is why $CD = 10.6$ in the first column of Table 2 is informative but not by itself a measure of how large the bias is.

3.3 Common correlated effects

PAPER The CCE estimator of Pesaran (2006) replaces the unobserved factors by cross-sectional averages of the data. With a lagged dependent variable the paper follows Chudik and Pesaran (2015) and adds p lags of the averages:

$$y_{i,t} = \rho_i y_{i,t-1} + x_{i,t}\beta_i + g_t\delta_i + \sum_{l=0}^{p} \bar{R}_{t-l}\gamma_{i,l} + e_{i,t}, \qquad \bar{R}_t = \frac{1}{N}\sum_{i=1}^{N} [\,y_{i,t},\; x_{i,t}\,]. \tag{4}$$

Averaging Equation (2) and the outcome equation over i gives $\bar{R}_t \approx \bar{\Gamma} f_t + (\text{observed factor terms}) + O_p(N^{-1/2})$. If $\bar{\Gamma}$ has full column rank, which requires at least as many averaged variables as unobserved factors, then f_t lies in the span of $\bar{R}_t$ up to vanishing error and including $\bar{R}_t$ removes the factor structure. The paper states consistency as $N, T \to \infty$ with $T/N \to 0$. The country slopes are averaged by the mean group estimator, with the variance estimated from the dispersion of country estimates:

$$\hat{\beta}_{MG} = \frac{1}{N}\sum_{i=1}^{N} \hat{\beta}_i, \qquad \widehat{\mathrm{Var}}(\hat{\beta}_{MG}) = \frac{1}{N(N-1)}\sum_{i=1}^{N} (\hat{\beta}_i - \hat{\beta}_{MG})(\hat{\beta}_i - \hat{\beta}_{MG})'. \tag{5}$$

EVIDENCE The effective sample for the standard errors is $N = 30$ countries, not 2,160 observations, which sets the power of every group comparison made later.

3.4 Reduced form with observed and unobserved factors

PAPER The authors then write the observed global factors as themselves exposed to the unobserved ones, $g_t = \xi f_t + \omega_t$, and the regressors as $x_{i,t} = \gamma_i^x f_t + \theta_i^x g_t + \chi_{i,t}$. Substituting into the outcome equation gives (reviewer's rendering of the paper's Equation 9, whose HTML typesetting is garbled):

$$y_{i,t} = \rho_i y_{i,t-1} + \beta_i \chi_{i,t} + (\beta_i \theta_i^x + \theta_i)\omega_t + [\beta_i(\gamma_i^x + \theta_i^x \xi) + \gamma_i + \theta_i \xi] f_t + \varepsilon_{i,t}. \tag{6}$$

The derivation is a direct substitution: replace $x_{i,t}$ by its factor representation, replace g_t by $\xi f_t + \omega_t$ everywhere, and collect the coefficients of $\chi_{i,t}$, ω_t and f_t. Equation (6) shows the identification logic. The country specific idiosyncratic variation in x (that is $\chi_{i,t}$) identifies the pure domestic effect β_i. The coefficient on f_t mixes four channels (direct loading, loading through x, and loading through g twice), so it cannot be separated into economically named effects, which is precisely why the authors resort to principal components of the composite term.

3.5 Three-step estimation and the interaction with the regime

PAPER Step 1 estimates Equation (4) by CCE-MG and computes the composite residual $\hat{u}_{i,t}$, which contains the observed and unobserved factor components. Step 2 extracts principal components PC_t from the $T \times N$ matrix $\hat{u}_{i,t}$, with their number chosen by the Ahn and Horenstein (2013) growth ratio. Step 3 replaces the factors by the components:

$$y_{i,t} = \rho_i y_{i,t-1} + x_{i,t}\beta_i + PC_t \kappa_i + \zeta_{i,t}. \tag{7}$$

INFERENCE The regime comparison reported in column (7) of Table 2 is, from the text and the table layout, the version of Equation (7) in which the component loadings are split by an indicator F_i for a strictly fixed regime, namely $PC_t \kappa_i^0 (1 - F_i) + PC_t \kappa_i^1 F_i$, and the mean group coefficients $\bar{\kappa}^1$ and $\bar{\kappa}^0$ are compared. The paper does not write this interaction equation explicitly, so this specification is the reviewer's reconstruction. The factor number estimator is

$$GR(k) = \frac{\ln(1 + \mu_k / V(k))}{\ln(1 + \mu_{k+1} / V(k+1))}, \qquad V(k) = \sum_{j>k} \mu_j, \qquad \hat{M} = \arg\max_k GR(k), \tag{8}$$

where μ_k is the k-th eigenvalue of the covariance matrix of the panel. EVIDENCE Table 10 reports $\hat{M} = 3$ for the non-core ratio over 2004 to 2022 and for the loan to deposit ratio $\hat{M} = 5$, and seven for the eight observed global variables taken together, so the observed list is close to its own rank limit.

4 Data and Empirical Strategy

4.1 Data

PAPER The data are a quarterly panel from 2004:Q1 to 2022:Q1 for high income economies. The full list has 31 countries and 2,263 country-quarters (Table 1). The benchmark drops Iceland, an outlier, and the first quarter because of the lag, leaving 30 countries and 72 quarters, hence 2,160 observations (Table 2). The non-core ratio is IFS line 26c (foreign liabilities of other depository corporations) divided by IFS lines 24 and 25 (transferable and other deposits). The loan to deposit ratio is IFS line 22d over the same denominator. Missing values are linearly interpolated, following Miranda-Agrippino and Rey (2020), and Bank of Canada and Bank of England balance sheets are hand matched for the two countries where IFS coverage is thin. Domestic controls are winsorised at the 1st and 99th percentiles. Country variables come from the IFS, the OECD, the Federal Reserve Bank of St. Louis, and the IMF integrated Macroprudential Policy database (Alam et al. 2019). The regime indicator comes from the coarse classification of Ilzetzki, Reinhart and Rogoff (2019), with end of quarter values, and is assumed not to change after 2019 because coverage ends there.

Table 1. **Summary statistics for selected variables (reproduced from Table 1 of the paper)**

Variable	Unit	Obs.	25th pct.	Mean	75th pct.
Non-core ratio	%	2,263	22.09	78.00	94.75
Loans to deposits	%	2,263	87.38	122.03	142.32
Capital to assets	%	2,263	5.39	7.22	8.30
BIS cross-border liabilities	%	1,656	29.29	84.09	110.62
BIS liabilities to banks	%	1,579	18.39	53.98	69.71
BIS liabilities to nonbanks	%	1,656	6.14	20.99	28.57
BIS liabilities to other sectors	%	1,579	0.00	9.06	11.93
GDP growth	%	2,263	-2.69	0.57	4.33
Real interest rate	pp	2,263	-2.30	-0.28	1.79
VIX (log)	index	2,263	2.62	2.87	3.05
US shadow rate	%	2,263	-1.20	0.69	2.03

Source: copied from the paper (official). Percentiles as printed by the authors. The text reports standard deviations of about 94 and 58 percent for the first two rows.

The mean non-core ratio of 78 percent means that foreign liabilities are on average about three quarters of deposits, which is large and dominated by financial centres. The inter-quartile range, 22 to 95 percent, shows a skewed distribution, which is relevant because mean group inference averages country coefficients without trimming. The BIS locational banking statistics cover 23 of the 30 countries and supply the currency and counterparty decomposition used in Section 5.4.

4.2 Empirical specification and identification

PAPER The identifying variation has two sources. For the domestic coefficients β_i it is the country specific idiosyncratic variation that remains after the factors are projected out. For the regime effect it is the cross-sectional difference in factor loadings between 16 pegged and 14 floating economies, with the factors themselves common to everyone. CRITIQUE The regime variable is a country characteristic that is, in this sample, time invariant. It identifies a between-group contrast in loadings, not a within-country change in regime. No event, policy break, or instrument isolates an exogenous change in regime. The design can therefore establish that fixed and floating groups differ in factor sensitivity, but a causal reading requires the assumption that group membership is unrelated to anything else that affects sensitivity.

Table 8 of the paper allows a check of the composition of the groups. The 16 fixed economies are Austria, Belgium, Cyprus, Denmark, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, Slovenia and Spain. Of these, 15 are euro area members, and Denmark is the only peg outside the euro. EVIDENCE The floating group contains Australia, Canada, Chile, the Czech Republic, Hungary, Israel, Japan, Korea, Norway, Poland, Romania, Sweden, the United Kingdom and the United States, and the macroprudential split is analysed in Section 5.3.

4.3 Data limitations and biases

- Regime composition: fixed means euro area membership in almost every case, so currency union, shared supervisor, and intra-area interbank integration are bundled with the exchange rate regime.
- Measurement: foreign liabilities in the numerator of the non-core ratio are not necessarily exposed to the exchange rate; for monetary union members much of the foreign funding is denominated in the domestic currency and may be provided by other area banks. The BIS split in Section 5.2 of the paper supports this, since the share in domestic and foreign currency moves similarly.
- Interpolation and hand matching: linear interpolation of missing IFS observations and stitching of different capital ratio series for Canada and the euro area introduce smoothing and potential breaks, and the stronger persistence it creates interacts with the lagged dependent variable.
- Look-ahead in regime coding: the classification is held fixed after 2019, and the group variable is time invariant, so countries such as Estonia, Slovenia, Greece and Cyprus are treated as fixers throughout, including the years before euro adoption.
- Short sample: only one full global cycle (2008 to 2012) and the pandemic are in the data, and the post-2015 period contains the policy regime change in the euro area. Data mining is limited by the pre-specified controls, but the choice of 8 observed factors and a binary macroprudential split is not tested for sensitivity.

5 Results

5.1 Benchmark estimates and the cross-sectional dependence diagnostic

EVIDENCE Table 2 reproduces the columns of the paper's Table 2 that carry the argument. The question it answers is whether the estimated effect of domestic variables and of global factors on the non-core ratio changes once cross-sectional dependence is handled. The dependent variable is the non-core ratio in percent. Column (1) is two-way fixed effects with a homogeneous slope; columns (4) and (5) are mean group estimators without and with cross-sectional averages; columns (6) and (7) add the principal components, with column (7) splitting their loadings by regime.

Table 2. **Benchmark regressions of the non-core ratio, 30 countries, 2004:Q1 to 2022:Q1 (selected columns of Table 2 of the paper)**

Regressor	(1) TWFE	(4) MG	(5) MG-CCE	(6) MG, PC	(7) MG, PC by regime
Lagged dependent variable	0.986[a]	0.714[a]	0.541[a]	0.659[a]	0.659[a]
	(0.016)	(0.030)	(0.035)	(0.031)	(0.031)
Real effective exchange rate	0.098[a]	0.700[b]	0.664[a]	0.424[b]	0.424[b]
	(0.025)	(0.284)	(0.256)	(0.187)	(0.187)
Money to GDP	0.046[a]	-0.022	0.252	0.175[a]	0.175[a]
	(0.017)	(0.051)	(0.186)	(0.064)	(0.064)
PC1, floating (fixed equal to 0)				1.297[b]	0.304[c]
				(0.537)	(0.164)
PC2, floating				0.703[c]	-0.082
				(0.398)	(0.157)
PC3, floating				1.197[b]	0.374[c]
				(0.608)	(0.204)
PC1, fixed (fixed equal to 1)					0.993[c]
					(0.531)
PC2, fixed					0.785[b]
					(0.359)
PC3, fixed					0.823
					(0.591)
CD statistic (p value)	10.59 (0.000)	12.76 (0.000)	0.80 (0.425)	-0.09 (0.929)	-0.09 (0.929)
R^2 (within)	0.911	0.960	0.053	0.958	0.958

Source: copied from the paper (official); standard errors in parentheses. Significance: a 1 percent, b 5 percent, c 10 percent. Observations 2,160 in every column. Capital regulation, liquidity regulation, capital to asset ratio, GDP growth and the real interest rate are omitted for space and are insignificant in most columns. Layout adapted by the reviewer.

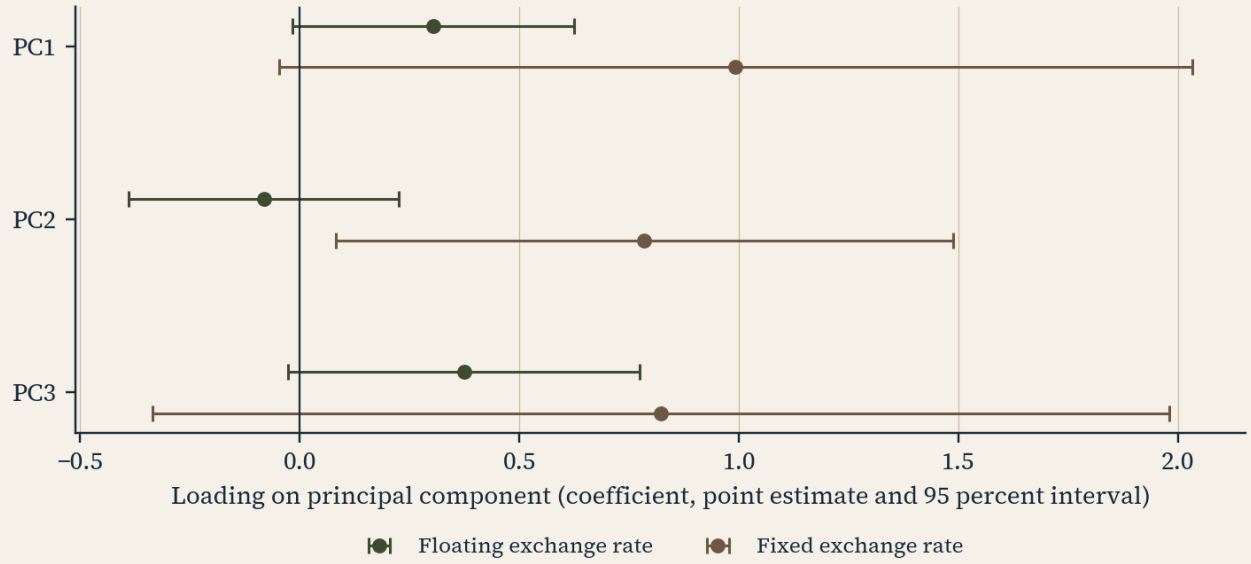
The first message of the table is that the lagged dependent variable drops from 0.986 in the pooled estimator to 0.659 in the mean group regressions. In dynamic heterogeneous panels, pooled estimators are biased towards one because country intercepts and slopes are forced to be equal (Pesaran and Smith 1995); the half-life of a shock implied by the pooled coefficient is $\ln 0.5 / \ln 0.986 \approx 49$ quarters, against $\ln 0.5 / \ln 0.659 \approx 1.7$ quarters in column (6) (reviewer's calculation). This is a very large difference in the dynamics the authors claim to explain, and it shows that the pooled estimates describe a series close to a unit root while the mean group estimates describe a fast mean reverting series. The second message is that the CD statistic is decisive: it

rejects weak dependence in columns (1) to (4), and is not rejected only once the cross-sectional averages or the components are included. The third message is that, with the factor structure removed, most domestic variables are insignificant, apart from the real exchange rate and money to GDP. A one unit rise in the real effective exchange rate index (base 100 in 2010) is associated with a short-run rise in the non-core ratio of about 0.42 points and a long-run rise of $0.424/(1-0.659) \approx 1.24$ points (reviewer's calculation using the mean coefficients, which is only an approximation because the mean of ratios is not the ratio of means). Appreciation raises foreign funding, consistent with the interest rate and valuation channels in Bruno and Shin (2015).

5.2 Global factors and the exchange rate regime

EVIDENCE In column (6), all three components enter positively and are significant at 5 or 10 percent, with loadings of 1.30, 0.70 and 1.20 for floating economies and effects measured per unit of each component. Component scaling is not reported, so a coefficient cannot be converted to percentage points of the non-core ratio, which has a standard deviation of about 94 points in the pooled data (Table 1). Column (7) splits the loadings by regime.

Figure 1. **Loadings of the non-core ratio on the three principal components, floating versus fixed exchange rate regimes, 2004:Q1 to 2022:Q1**



Source: point estimates and standard errors from Table 2, column (7) of the paper (official numbers); chart generated by the reviewer. Intervals are plus or minus 1.96 standard errors. Component units are not reported by the authors.

Figure 1 plots column (7). For the first component the fixed regime point estimate is 0.99 against 0.30 for floaters, for the second 0.79 against -0.08, for the third 0.82 against 0.37. All three fixed estimates lie above the floating ones, which is the sign the trilemma predicts, but the intervals overlap for the first and third components. The figure is correlational: it compares group averages of country loadings, and the variation that drives it is the cross-sectional difference between 16 and 14 countries.

To judge how sharp the contrasts are, the reviewer computed the differences between the two groups from the reported coefficients and standard errors, treating the two group estimates as independent because the mean group averages are taken over disjoint sets of countries.

Table 3. **Differences in component loadings between fixed and floating regimes (reviewer's calculation from Table 2, column 7)**

Component	Fixed minus floating	Standard error	t ratio	Verdict at 10 percent
PC1	0.689	0.556	1.24	Not significant
PC2	0.867	0.392	2.21	Significant
PC3	0.449	0.625	0.72	Not significant
Sum of three loadings	2.005	0.924	2.17	Significant (approximation)

Source: generated by the reviewer from the numbers in the paper. The standard error of the sum assumes zero covariance between the three component loadings within a group, which the paper does not report; the result is therefore approximate.

Only the second component differs individually. The aggregate contrast supports the trilemma reading, with a t ratio of about 2.2 and a two-sided normal p value of about 0.03 on the reviewer's approximation. The paper reports t equal to 5.18 with p equal to 0.011 for this test (Table 3, column 7), a pair that cannot come from a standard normal or large-sample t distribution, since a t of 5.18 corresponds to a p value below 0.001. The discrepancy might reflect a different test statistic or a typographical problem in the HTML version, and cannot be resolved from the paper. CRITIQUE A further point concerns the sum itself. Principal components are identified only up to sign, so a statistic built on the algebraic sum of component loadings changes if the sign of one component is flipped; here all three differences happen to be positive, which makes the sum look natural, but the test is not invariant to the normalisation.

The long-run implication is more tangible. With the mean persistence of 0.659 the multiplier is $1/(1-0.659) = 2.93$, so the long-run loadings are $0.993 \times 2.93 \approx 2.9$ for fixers and $0.304 \times 2.93 \approx 0.9$ for floaters on the first component (reviewer's approximation, same caveat as above). A fixed regime thus amplifies the first component by a factor of about three, which is economically large if it is real, though the difference is not statistically sharp.

5.3 Macroprudential policy and the regime split

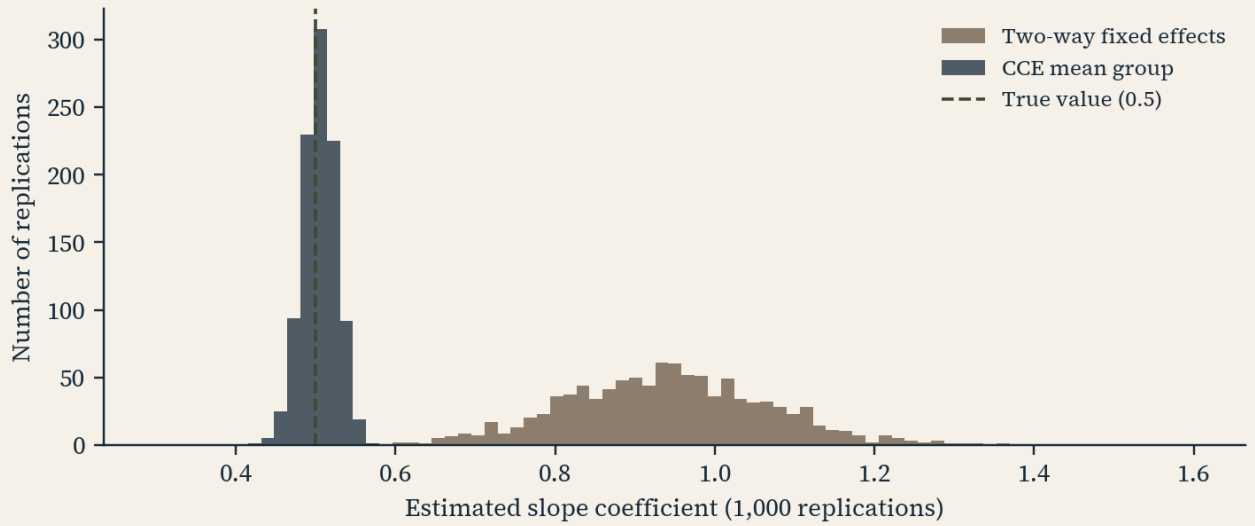
EVIDENCE Table 5 of the paper splits economies by whether they tightened macroprudential policy more than the median country over the sample. The fixed regime countries divide into 6 tight and 10 loose economies and the floaters into 8 tight and 6 loose (reviewer's count from Table 8, Iceland excluded). Among fixers the loading on PC2 is 1.457 (standard error 0.519) for loose and 0.015 (0.366) for tight economies, and on PC3 it is 1.558 (1.009) and -0.015 (0.415). The reviewer's difference for PC2 is 1.44 with a standard error of 0.64, a t ratio of 2.27, and for PC3 it is 1.57 with a standard error of 1.09, a t ratio of 1.44. For floaters the corresponding loadings show no consistent prudential pattern, as the authors state. The first component loads more, not less, on tight economies (0.884 against 0.413 in the full sample), which the authors explain by the fact that PC1 follows world macroprudential tightening and tight countries follow the world trend more closely.

5.4 Interpreting the components and the funding channel

EVIDENCE Table 6 regresses each component on the standardised observed global variables. PC1 has $R^2 = 0.971$ and loads mainly on world macroprudential tightness (coefficient -2.95, standard error 0.21) and the shadow rate (1.09, standard error 0.10). PC2 has $R^2 = 0.920$ and loads on the shadow rate (1.17) and, more weakly, on the non-bank share (0.64, the opposite sign to PC1) and on world macroprudential tightness (-0.59). PC3 has $R^2 = 0.675$, rising to 0.786 when crisis dummies are added, where the dummy for the 2007:Q4 to 2011:Q4 banking crisis has a coefficient of 2.04 (standard error 0.46). The authors summarise PC1 as macroprudential, PC2 as the US shadow rate and PC3 as crises.

Table 7 regresses BIS cross-border liabilities scaled by deposits on the components with country fixed effects. The coefficients on the three components for total liabilities are 0.196, 0.250 and 0.400, with those for liabilities to banks of 0.198, 0.218 and 0.392 and for nonbank financials of 0.017, 0.049 and 0.015 (insignificant). Liabilities to other sectors show negative coefficients on PC1 and PC2. The foreign currency and domestic currency columns are close to each other (0.100 and 0.109 for PC1). EVIDENCE The channel is therefore the interbank market, and the currency of the liability is not a discriminating feature. The persistence in these regressions is about 0.958, so the implied long-run multiplier is close to 24 and the contemporaneous coefficients are small relative to it; these standard fixed effects regressions do not address the heterogeneity and dependence the paper stresses elsewhere.

Figure 2. Illustrative bias of two-way fixed effects against a CCE mean group estimator when country loadings on a common factor are correlated with the regressor loadings



Source: simulation generated by the reviewer, not from the paper. Design: 30 countries, 72 quarters, one AR(1) factor with coefficient 0.9, regressor loadings with mean 1 and standard deviation 0.7, outcome loadings equal to one plus 0.8 times the deviation of the regressor loading plus noise of standard deviation 0.5, true slope 0.5, 1,000 replications.

6 Robustness and Additional Tests

EVIDENCE The paper reports Table 4 (robustness of column 7), Table 5 (macroprudential split), Table 6 and 7 (mechanisms), Table 10 (factor numbers) and the loan to deposit decomposition of Table 11. The assessment below is a reviewer's judgement of whether each test addresses the concern it is aimed at.

Table 4. Robustness tests: concern, test, result and assessment

Concern	Test reported	Result	Does it resolve the concern?
US dominance of the observed factors	Exclude the United States (Table 4, col. 1)	Fixed loadings 1.18, 0.75, 0.85; floating 0.31, -0.09, 0.41; pattern unchanged	Partly. Dropping one of 14 floaters leaves the shadow rate and the dollar as global drivers, so the test does not remove the US centred nature of the factors.
Outlier country	Include Iceland (col. 2)	Estimates less precise; CD statistic 2.93 rejects weak dependence	No. The result is sensitive to one country, which indicates fragility of the factor extraction.
Pandemic episode	Sample ends in 2019:Q4 (col. 3)	Almost unchanged, lower significance	Yes for the pandemic, with small loss of power.
Global financial crisis	Sample 2010:Q1 to 2019:Q4 (col. 4)	Fixed PC1 1.66 against 0.41 for floaters; PC3 insignificant	Partly. The gap in PC1 is larger but not shown to be significant (reviewer t about 1.5), and the sample is 40 quarters.
Outcome variable definition	Loan to deposit ratio (col. 5)	Floating PC1 0.67 exceeds fixed 0.39; insulation absent in the full sample	No. The headline contrast reverses for a closely related variable; the authors' explanation relies on unreported regressions.
Estimator choice	Columns (1) to (5) of Table 2	CD rejects in (1) to (4); not in (5) to (7)	Yes for the need to model dependence; not a test of the regime contrast itself.
Slope heterogeneity	Delta tests, Table 3	Homogeneity rejected	Yes, for allowing country slopes; with $N = 30$ the group averages remain imprecise.
Meaning of the components	Regressions on observed factors, Tables 6 and 11	High R^2 with macroprudential index, shadow rate and crisis dummies	Partly. High R^2 between trending series does not establish an economic interpretation.
Currency and counterparty	BIS decomposition, Tables 7 and 12	Interbank liabilities drive the effect; currency irrelevant	Partly. It locates the channel but is a standard fixed effects regression on 23 countries.

Source: constructed by the reviewer from Tables 2 to 7, 10 and 11 of the paper; numbers copied from the paper unless marked as reviewer calculation.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Two-way fixed effects leave strong cross-sectional dependence in the residuals of the non-core ratio equation (CD of 10.6 in the first column), so the common factor structure cannot be ignored. The mean group estimates with cross-sectional averages remove the dependence, and country slopes are heterogeneous. Persistence is much lower once heterogeneity is allowed. Foreign interbank liabilities, not nonbank or other sector liabilities, carry the co-movement with the components.

SUPPORTED BUT CONDITIONAL

Fixed regime banking systems load more on the global components than floating ones. The sign is consistent across all three components in the benchmark, the aggregate difference is about two standard errors on the reviewer's approximation, and only PC2 differs individually. The conclusion depends on the regime coding, on 30 countries, and on the particular components extracted.

SUGGESTIVE

That insulation is stronger away from crises (support is a larger point estimate that is not tested). That tighter macroprudential policy complements the exchange rate regime for fixers, where groups contain 6 and 10 countries and the classification is itself an outcome of past conditions. That PC1 is a macroprudential factor and PC2 a shadow rate factor.

NOT ESTABLISHED

That fluctuations in the non-core ratio are mainly driven by global rather than domestic factors in a variance decomposition sense, that the exchange rate regime itself, as distinct from euro area membership, causes the insulation, and that the macroprudential effect is causal.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- A treatment of cross-sectional dependence and heterogeneity that earlier trilemma panels lacked, with explicit diagnostics (CD statistic, factor number estimator, slope heterogeneity tests) after every step.
- A composite view of the global financial cycle through principal components, rather than a single proxy such as the VIX, which acknowledges that no single observed variable spans the factor space.

- A balance sheet outcome with a documented link to credit booms and crises (Jordà, Schularick and Taylor 2017; Hahm, Shin and Shin 2013), and a decomposition by currency and counterparty from BIS data.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Transparent reporting of unfavourable results, for example the Iceland column and the loan to deposit column, which allows the reader to see where the headline fails.
- Use of a restricted sample of advanced economies with open capital accounts (Chinn and Ito mean 0.93), which isolates the regime from capital controls, a clean design choice.
- Computationally replicable: the authors use public Stata routines (xtdcce2, xtnumfac) and public series, so replication is feasible.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The fixed regime group is, with one exception, the euro area. Of 16 fixed economies in Table 8, 15 are euro members and the sixteenth is Denmark. **Why it matters.** The fixed versus floating contrast is thus a contrast between a monetary union with a single currency, integrated interbank markets and common supervision, and a heterogeneous set of floaters, which includes the United States whose shadow rate is one of the global factors. The authors' finding that the channel is interbank borrowing in both domestic and foreign currency is what intra-area wholesale funding would produce independently of any exchange rate insulation mechanism. **Potential bias.** The estimated regime effect would overstate the effect of the exchange rate regime on exposure to global funding conditions, since it also captures the degree of cross-border financial integration, which is higher inside the area. **How it could be tested.** Exclude the euro area from the fixed group and compare the non-euro pegs (Denmark, and Estonia before its euro adoption in 2011) with euro members; re-measure the numerator net of liabilities to other euro area residents; or add a euro membership indicator interacted with the components alongside the regime indicator.

OTHER WEAKNESSES

- Variance claim not shown. The text states that fluctuations are overwhelmingly determined by global factors, yet the statistic reported is the share of explained common variance carried by the components (79.5 percent), and the R^2 values of 0.95 in the mean group regressions are dominated by the lagged dependent variable. Why it matters: the headline conclusion is not a variance decomposition of the non-core ratio. Bias: overstated global dominance. Test: variance decomposition of the change in the ratio into common and idiosyncratic components, with and without the lag.
- Generated regressors. The components are estimated from the same panel and then used as regressors in a second regression, without any adjustment of standard errors for the estimation

error. Why it matters: significance can be overstated, in particular at the 10 percent level where many results sit. Test: bootstrap over the three steps, with resampling across countries.

- Trending series and interpretation. PC1 regressed on eight smooth global series gives $R^2 = 0.97$ with 72 observations. Trending or near unit root series generate high R^2 without any relation. The non-core ratio itself falls from 70 to 30 percent between 2008 and 2022, and world macroprudential tightness rises. Test: first differences, cointegration tests, or detrending, and a placebo using shuffled or lagged world series.
- Failure of the stated asymptotic condition. The paper states consistency when $N, T \to \infty$ and $T/N \to 0$. In this panel $T/N = 72/30 = 2.4$. Why it matters: the finite sample behaviour of the CCE-MG estimator with $N = 30$ is the relevant one. Test: Monte Carlo calibrated to the panel dimensions (the illustration in Figure 2 uses a simpler static design).
- Endogenous classification. Macroprudential tightness is classified by cumulative tightening over the sample, which depends on how exposed the country was to the funding booms and crises the paper studies. Reverse causality can create the apparent protective effect. Test: use the pre-sample level of tightness or an exogenous change in rules.
- Sign non-invariance and rotation. The sum of component loadings is not invariant to the sign of components, and the economic interpretation of PC2 (a positive shadow rate coefficient contradicting Miranda-Agrippino and Rey 2020) rests on a post hoc argument about quantitative easing without a direct test.
- External validity. Thirty advanced economies in one global cycle say little about emerging markets, where the trilemma debate (Obstfeld, Ostry and Qureshi 2019) was originally contested.

8 Contradictions and Edge Cases

CRITIQUE **When should the result disappear?** If the fixed group were reclassified so that euro members were treated separately, or if the numerator excluded intra-area liabilities, the regime contrast could vanish. The authors give no evidence against that possibility, and the BIS result that domestic and foreign currency liabilities respond similarly is what intra-area funding implies.

CRITIQUE **Is the factor estimator reliable at these dimensions?** The illustration in Figure 2 uses a simple static design with one factor whose loadings are correlated across the outcome and the regressor. The two-way fixed effects estimate of a true slope of 0.5 averages 0.94 across 1,000 replications and the pooled within-country estimate averages 1.23, while the CCE mean group estimate averages 0.505 with a standard deviation of 0.021. The simulation is generated by the reviewer and illustrates the omitted variable bias that motivates the paper; it does not test the authors' dynamic three-step procedure with 30 countries and heterogeneous persistence, for which a bespoke Monte Carlo would be needed.

CRITIQUE **Parameter dependence.** The lagged dependent variable differs sharply by estimator (0.986 against 0.659). If true persistence were near the pooled value the long-run effects would not be 2.9 and 0.9 but several times larger; if it is as low as 0.54 (column 5), the multiplier falls to 2.2. Conclusions

about the size of the regime effect therefore depend on a coefficient that moves by a third to a half across specifications.

CRITIQUE **Extreme values and boundary cases.** If all countries were fixed or all floating, the contrast would be unidentified; with 16 against 14 it is identified, but a single change in classification moves a group mean by about 6 percent of its weight, which matters when one country has a large loading. The Iceland column shows that one country can alter the factor structure and reintroduce cross-sectional dependence. Outside the sample, in economies with capital controls or with dominant dollar liabilities, the mechanism (foreign interbank funding responding to the stance abroad) could run in the opposite direction, because floating currencies with foreign currency debt face a valuation channel that a peg removes.

CRITIQUE **Alternative mechanisms.** The same pattern could arise from (a) the euro area sovereign debt crisis of 2010 to 2013, which hit fixers in the group and reduced interbank funding, captured in the paper by the crisis component but not separated from regime; (b) a global deleveraging of European banks after 2008 that is a banking-group shock, not a currency-regime effect; and (c) dollar funding stress, which the paper handles only through the US real effective exchange rate.

9 Further Literature

Table 5. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Rey (2015)	A global financial cycle drives national credit and asset prices, turning the trilemma into a dilemma	Foundation; the present paper finds partial insulation by floating rates, a softer reading	Defines the hypothesis being tested
Miranda-Agrippino and Rey (2020)	US monetary policy as the source of a global factor in asset prices and credit	Preceded; PC2 has a positive shadow rate coefficient, opposite to their sign, and the authors explain it by quantitative easing	Shows the identification of the global factor is not settled
Bruno and Shin (2015)	Global bank leverage, dollar valuation and cross-border funding	Theoretical foundation of the non-core mechanism	Supplies the economic channel that the paper tests only reduced form
Shin and Shin (2011); Hahm, Shin and Shin (2013)	Non-core liabilities as a measure of financial vulnerability	Established the outcome variable	The ratio studied is their indicator, adapted to foreign liabilities
Obstfeld, Shambaugh and Taylor (2005)	Interest rate pass-through by exchange rate regime over a century	Preceded; methodology uses rates, not balance sheets	The paper translates the trilemma from prices to quantities
Obstfeld, Ostry and Qureshi (2019)	Flexible rates dampen sensitivity of emerging market rates to global conditions	Similar finding in another sample and variable	Supports the direction of the regime result
Klein and Shambaugh (2015)	Partial pegs and capital controls and monetary autonomy	Related; regime coding beyond a binary peg	Suggests a finer regime measure than the strict peg dummy

Paper	Shared idea	Relationship	Why it matters
Cerutti, Claessens and Rose (2019)	Share of capital flow variation explained by a global factor	Contradicts the magnitude: less than a quarter against overwhelming here	Differences in outcome and method need reconciliation
Amiti, McGuire and Weinstein (2017)	Global factor in cross-border banking during booms only	Related; time dependence of global importance	Consistent with the stronger insulation outside crises
Pesaran (2006); Chudik and Pesaran (2015)	Common correlated effects for heterogeneous panels, dynamic extension	Methodological foundation	Justifies step 1 and the use of cross-sectional averages
Pesaran and Smith (1995)	Pooled dynamic estimators are biased under heterogeneity	Explains the fall of persistence from 0.986 to 0.659	Shows why the estimator choice changes the dynamics
Bai (2009); Ahn and Horenstein (2013)	Interactive fixed effects and the number of factors	Methodological alternatives	Principal components and factor number rely on these
di Giovanni, Kalemli-Ozcan, Ulu and Baskaya (2021)	Global push of financial conditions on local credit, with firm level data	Improved identification of the local credit effect	Shows a path to causal estimates the paper lacks
Jordà, Schularick and Taylor (2017)	Bank balance sheet ratios as crisis predictors	Motivates the outcome	Gives the stability relevance of the non-core ratio

Source: constructed by the reviewer. Entries cited in the paper's bibliography were checked against it; Klein and Shambaugh (2015) and others are listed from the reviewer's knowledge and were not opened for this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** The regime effect is not separated from monetary union. *Why it matters:* It is the paper's central claim. *Existing limitation:* 15 of 16 fixers are euro members. *Way forward:* event studies around euro adoption and a net-of-area measure of foreign liabilities.
- **Gap:** No variance decomposition of the non-core ratio and no test of the difference between regime differences (full sample against non-crisis sample). *Why it matters:* Both are headline claims. *Existing limitation:* point estimates and sum tests only. *Way forward:* a joint test of the interaction of regime, crisis and component, with bootstrapped standard errors.
- **Gap:** The economic identity of the components. *Why it matters:* Policy use requires named factors. *Existing limitation:* correlations between trending series. *Way forward:* identified shocks, for example high frequency monetary surprises, as in Miranda-Agrippino and Rey (2020).

10.2 Literature-level gaps

- **Gap:** Reconciliation of large and small estimates of the global factor share (Cerutti, Claessens and Rose 2019 against this paper). *Why it matters:* Determines how much room domestic policy has. *Existing limitation:* different outcomes, samples and factor definitions. *Way forward:* apply identical decomposition to flows, stocks and balance sheet ratios in one panel.
- **Gap:** Distinguishing insulation of prices from insulation of quantities. *Why it matters:* Rates may be insulated while quantities and leverage are not (Bruno and Shin 2015). *Existing limitation:* the literature tests one at a time. *Way forward:* joint estimation of rate and funding ratio responses.

10.3 Methodological gaps

- **Gap:** Inference after generated regressors in multi-step factor estimation. *Why it matters:* Many results sit at the 10 percent level. *Existing limitation:* no correction in the paper. *Way forward:* panel bootstrap or a one-step Bayesian factor model.
- **Gap:** Small number of countries with heterogeneous slopes. *Why it matters:* Mean group inference with $N = 30$ is imprecise. *Existing limitation:* no shrinkage. *Way forward:* hierarchical models with partial pooling of country loadings.

11 Research Opportunities

EXTENSION The ideas below are the reviewer's, not the authors'. Data sources are named as candidates and their availability and coverage must be checked at implementation.

Idea 1. Separating monetary union from the exchange rate regime with euro adoption events

RESEARCH QUESTION. Does adopting the euro change a banking system's loading on global factors, after the exchange rate regime has been fixed by a prior peg?

MOTIVATION. In this paper the fixer group is 15 euro members out of 16, so the regime effect cannot be distinguished from union effects.

HYPOTHESIS. Loadings on global funding factors rise after adoption because intra-area wholesale funding becomes frictionless, independently of the exchange rate regime.

ECONOMIC MECHANISM. Elimination of currency risk inside the area increases cross-border interbank funding, which is the channel the paper isolates.

LITERATURE CONNECTION. Catão, Ditzen and te Kaat; Shambaugh (2004); Obstfeld, Shambaugh and Taylor (2005).

DATA. IFS and ECB monetary financial institutions balance sheets, BIS locational banking statistics, adoption dates (Slovenia 2007, Cyprus and Malta 2008, Slovakia 2009, Estonia 2011, Latvia 2014, Lithuania 2015, Croatia 2023).

VARIABLES. Non-core ratio gross and net of liabilities to other euro area residents; adoption indicator; global factors.

METHOD. Staggered difference in differences with heterogeneous treatment effects, combined with CCE or interactive fixed effects.

IDENTIFICATION STRATEGY. Timing of adoption among countries already pegged, with pre-trend tests and the outcomes net and gross of area liabilities.

EXPECTED CONTRIBUTION. A clean estimate of the union effect that either supports or reverses the regime interpretation.

MAIN RISK. Few treated units and endogenous adoption timing (accession is tied to macro convergence).

Idea 2. Dollar funding, swap lines and the asymmetry of insulation

RESEARCH QUESTION. Is the insulation from a floating exchange rate limited to non-dollar funding, and do central bank swap lines reduce the loading on global factors?

MOTIVATION. The paper finds that currency denomination does not matter, but its test is a fixed effects regression on 23 countries and averages across currencies.

HYPOTHESIS. Banks funded in dollars load on a dollar funding factor irrespective of the domestic regime, and swap line announcements cut that loading.

ECONOMIC MECHANISM. Dollar funding stress and deviations from covered interest parity propagate to banks that borrow dollars, not through the domestic rate.

LITERATURE CONNECTION. Bruno and Shin (2015); Avdjiev, Bruno, Koch and Shin (2019).

DATA. BIS locational banking statistics by currency and counterparty, CIP deviations, dates of Federal Reserve swap line announcements (2008 and 2020).

VARIABLES. Cross-border liabilities by currency over deposits; dollar factor; announcement indicators.

METHOD. Event study of the change in dollar liabilities around announcements, with CCE controls for the common factors.

IDENTIFICATION STRATEGY. Announcement dates are plausibly exogenous to individual banking systems and vary in eligibility across countries.

EXPECTED CONTRIBUTION. Clarifies which part of the global factor is a dollar factor and when floating rates fail to insulate.

MAIN RISK. Announcements cluster with global stress, so confounding with the crisis factor is likely.

Idea 3. A one-step Bayesian factor model with regime specific loadings

RESEARCH QUESTION. What share of the variance of the non-core ratio is global, and how large is the regime difference in loadings once estimation uncertainty is carried through?

MOTIVATION. The paper's three steps ignore generated regressor uncertainty and use a sign dependent sum test.

HYPOTHESIS. The posterior probability that fixed loadings exceed floating ones is high for one factor and moderate for others, and the global variance share is lower than the paper suggests.

ECONOMIC MECHANISM. Same as the paper, now with partial pooling.

LITERATURE CONNECTION. Bai (2009); Pesaran (2006); Hoffman and Gelman style Hamiltonian samplers.

DATA. The paper's public panel (IFS, FRED, iMaPP, Ilzetzki et al.).

VARIABLES. Non-core ratio, domestic controls, regime indicator, factors.

METHOD. Hierarchical dynamic factor model with country loadings drawn from regime specific priors, estimated by Hamiltonian Monte Carlo.

IDENTIFICATION STRATEGY. Factor rotation fixed by a sign and ordering convention that is imposed on posteriors, with regime contrasts expressed through invariant quantities such as loading norms.

EXPECTED CONTRIBUTION. A rotation invariant inference and a variance decomposition missing from the paper.

MAIN RISK. Weak identification of factors with N equal to 30 and slow mixing.

Idea 4. The trilemma cube for banks: regime by capital account openness

RESEARCH QUESTION. In a wider sample, does the insulation from a flexible rate depend on how open the capital account is?

MOTIVATION. The paper restricts to open advanced economies, so it cannot test the third leg of the trilemma.

HYPOTHESIS. Insulation is largest under floating rates with open accounts and disappears with controls, since controls remove the channel.

ECONOMIC MECHANISM. Capital controls limit the cross-border interbank leg.

LITERATURE CONNECTION. Obstfeld, Ostry and Qureshi (2019); Klein and Shambaugh (2015); Chinn and Ito (2006).

DATA. IFS, BIS locational statistics, Chinn-Ito index, Ilzetzki et al. classification, 2004 to 2023.

VARIABLES. Non-core ratio, regime, openness, global factors.

METHOD. CCE-MG with triple split of loadings.

IDENTIFICATION STRATEGY. Within-country changes in openness around liberalisation episodes.

EXPECTED CONTRIBUTION. A bank balance sheet test of the full trilemma.

MAIN RISK. Poor data coverage and heterogeneity of factor structure in emerging markets.

Idea 5. Out-of-sample forecasting content of the funding components

RESEARCH QUESTION. Do real-time estimates of the three components forecast bank equity returns, credit spreads or systemic stress indicators out of sample?

MOTIVATION. The paper interprets the components but does not test whether they carry information that markets do not already price.

HYPOTHESIS. The first and third components forecast stress with a positive out-of-sample R^2 at horizons of one to four quarters.

ECONOMIC MECHANISM. Rising interbank funding ratios signal fragility.

LITERATURE CONNECTION. Jordà, Schularick and Taylor (2017); Hahm, Shin and Shin (2013).

DATA. Paper's panel, ECB composite indicator of systemic stress, bank equity indices.

VARIABLES. Expanding window components, future stress index, bank returns.

METHOD. Panel predictive regressions with expanding window CCE and out-of-sample R^2.

IDENTIFICATION STRATEGY. Strict real-time estimation to avoid look-ahead, which is also a prediction not a causal claim.

EXPECTED CONTRIBUTION. A financial data-science application with a practical test of usefulness.

MAIN RISK. Short sample with a single crisis and a high risk of overfitting.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Euro adoption events	4	4	5	4	5	5	27
2. Dollar funding and swap lines	4	3	4	4	5	5	25
3. Bayesian factor model	4	3	5	5	3	4	24
4. Trilemma cube	3	3	3	3	4	3	19
5. Out-of-sample forecasting	3	4	3	4	3	3	20

Source: reviewer's subjective scoring.

The three strongest are Ideas 1, 2 and 3. Idea 1 attacks the single most important weakness with public data and an established design, Idea 2 opens a mechanism the paper dismisses with a weak test and has plausibly exogenous events, and Idea 3 is the most feasible methodological improvement and uses the existing panel.

12 Learning Framework

12.1 Prerequisite concepts

- The Mundell-Fleming trilemma and the notion of monetary independence.
- Bank balance sheets: core (deposit) and non-core (wholesale, foreign) funding, leverage and procyclicality.
- Panel data with cross-sectional dependence: factor models, strong versus weak dependence, projection on cross-sectional averages.
- Principal components, eigenvalues and the identification of factors up to rotation and sign.

12.2 Core concepts learned

- Two-way fixed effects removes only a common mean; heterogeneous loadings leave dependence that biases estimates.
- Pooled dynamic estimators overstate persistence when slopes are heterogeneous.
- A group contrast in factor loadings is a different object from a causal effect of the group variable.

12.3 Important equations

Equation (1) is the interactive fixed effects model; Equation (3) is the CD statistic; Equation (4) is the dynamic CCE regression; Equation (6) shows how the observed and unobserved factors enter the reduced form; Equation (8) is the growth ratio factor number estimator.

12.4 Methods learned

Common correlated effects with mean group averaging, factor number estimation, principal components of a composite residual, slope heterogeneity testing, and the use of standardised regressions to interpret estimated factors.

12.5 Questions to test understanding

COMPREHENSION

1. What is the non-core ratio and why is it argued to rise with credit booms?
2. Why does two-way fixed effects fail to remove cross-sectional dependence in this paper?
3. What are the three components and which observed variables are they tied to?

ANALYSIS

1. Why does CCE need the number of cross-sectional averages to exceed the number of unobserved factors?
2. Why does the mean persistence fall from 0.986 to 0.659 when the estimator changes?
3. Why is a sum of principal component loadings not invariant to the normalisation of the components?

RESEARCH

1. How would the regime contrast change if euro members were removed from the fixed group?
2. Which shock could identify the effect of a floating rate causally?
3. How would a one-step Bayesian factor model change the inference?

12.6 Papers to read next

Miranda-Agrippino and Rey (2020); Bruno and Shin (2015); Pesaran (2006); Obstfeld, Ostry and Qureshi (2019); Chudik and Pesaran (2015).

13 Conclusion

13.1 One-sentence thesis

Advanced economy bank funding ratios share a few global factors, and the paper finds, with a careful panel method, that they bite harder under fixed exchange rates, but the fixed group is essentially the euro area, so the claim concerns monetary union at least as much as the exchange rate regime.

13.2 Three most important insights

- Ignoring heterogeneous loadings on common factors leaves strong cross-sectional dependence and distorts both persistence and slopes.
- The global component of the non-core ratio runs through interbank liabilities, with little role for the currency denomination.
- Group contrasts of loadings, once their uncertainty is computed, are mostly weaker than the headline suggests.

13.3 Strongest evidence

The CD diagnostics in Table 2, which show dependence rejected under two-way fixed effects and not rejected after the factor structure is modelled, together with the aggregate loading difference of about two standard errors.

13.4 Biggest weakness

The regime variable is confounded with euro area membership, so the exchange rate regime is not identified separately from monetary union.

13.5 Most important research gap

A causal estimate of what a flexible exchange rate does to the global sensitivity of bank funding, separate from union and integration effects.

13.6 Best research opportunity

Idea 1: euro adoption events with net and gross liability measures.

13.7 Three papers to read next

Miranda-Agrippino and Rey (2020); Bruno and Shin (2015); Pesaran (2006).

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